

Task 27: Validate Stack Sequences

Problem Description

Given two integer arrays pushed and popped each with distinct values, return true if this could have been the result of a sequence of push and pop operations on an initially empty stack, or false otherwise.

Java Solution

```
class Solution {
    public boolean validateStackSequences(int[] pushed, int[] popped) {
        Stack<Integer> stack = new Stack<>();
        int i = 0;

        for (int x : pushed) {
            stack.push(x);
            while (!stack.isEmpty() && stack.peek() == popped[i]) {
                stack.pop();
                i++;
            }
        }

        return stack.isEmpty();
    }
}
```